VF Asia Footwear Laboratory Testing Manual

Test Method for Measurement of the Strength of Top Lift or Stud attachment	Test method #: VF- F10 Page 1 of 5
Date Issued: May,2019 Revision #: 1 Reason for revision : Rectify the unit of measure from kg to kgf for strength of material.	Created by: Nila Babia Erazo

PURPOSE AND SCOPE: To measure the strength of top lift or stud attached on the shoe heel

REFERENCE: SATRA TM 108

EQUIPMENTS:

1. Tensile testing machine meeting the requirements of ASTM D 638.
2. Devices for clamping :
   - A metal bar of diameter 5.5± 0.5 mm and minimum length of 100 mm
   - A stirrup, of minimum width 80mm, capable of supporting the bar at both ends
   - A loop of steel wire of diameter 1.2 ± 0.4 mm and minimum length of 150 mm
3. Hand or electric drill with twist drill bits of diameter @ 1.5 ±0.1 mm and @ 7.0 ± 0.5 mm
4. Calipers capable of measuring up to 0.01mm. Digital type is preferred.

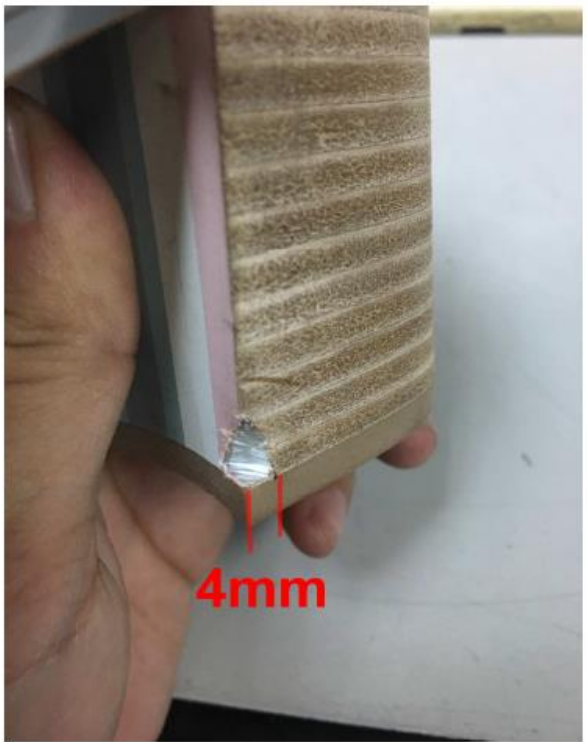
MATERIALS: Assembled shoes with top lift or stud on the heels or Heel with top lift or stud

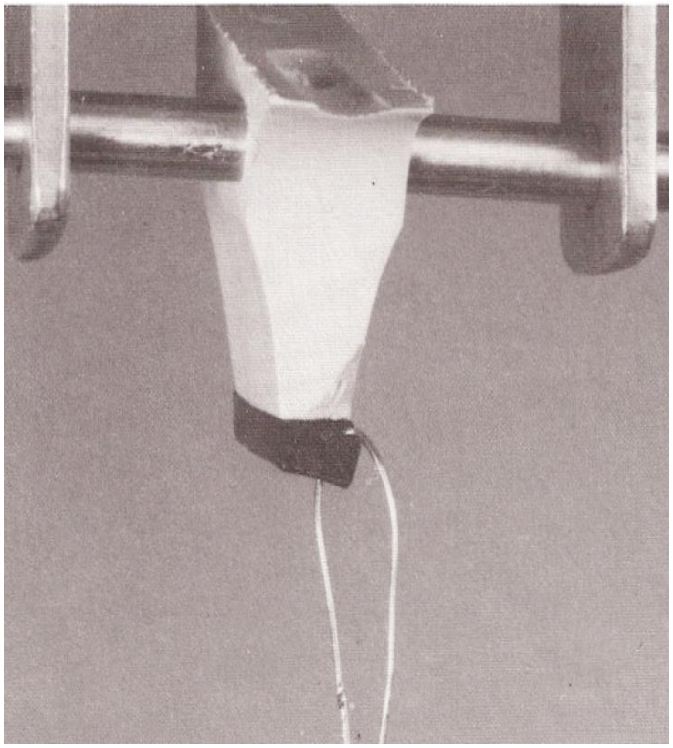
Construction A – Top lift without spigot
Construction B – Polymeric top lift with a spigot

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours prior to testing at the standard laboratory environment at 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.
2. If the heel is attached to completed footwear, remove the heel by extracting the heel attaching pins.
3. If the heel and top lift are supplied separately then the top lift must be attached by essentially the same method used in production.
   - ***Construction A – Top lift without spigot: It may not be possible to reproduce in the laboratory the attachment of "autolok "type top lift with a secondary stapled attachment***
   - ***Construction B – Polymeric top lift with a spigot-Spigot top lift should be inserted fully without twisting or straightening. The top lift attachment to the spigot can be assessed without the top piece being attached to the heel. In this case ignore the procedure 4 and 6.***
4. Grind (or cut using a hacksaw) a tapered flat on a breast edge corner of each heel immediately above the top lift to a depth of approximately 4-mm. This is to provide clearance for the wire loop to be threaded through the top lift, see ***Figure 1***.
   - ***Construction A – Top lift without spigot: Take care not to expose any secondary attachment devices such as staples or pints as this will reduce the overall top-piece attachment strength.***

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5. Use the drill bit to make a hole through the full thickness of each top lift, perpendicular to the top lift/heel interface and in line with the heel cut away. The hole should be approximately 2 mm from the edges of the top lift.
6. Use the drill bit to make a horizontal hole through the full thickness of the heel. **The position of this hole is critical and should be carefully chosen to ensure that the heel remains as close to vertical as possible during the test.** In general, this means the hole should be close to the prepared breast edge corner of the heel. If the heel tips during the test, then a significant component of force measured by the tensile testing machine will be parallel to the heel/ top lift interface which will give a result which is falsely high. ***See figures 1 and 2.***
    - Construction A – Top lift without spigot. The hole should be at least 10 mm above the top lift /heel interface to prevent drilling into the secondary attachment devices.
    - Construction B – Polymeric top lift with a spigot. Ensure that the hole is in a position that does not foul any hardened steel reinforcing pin and is through a sufficient amount of plastic to anchor the heel during the test.

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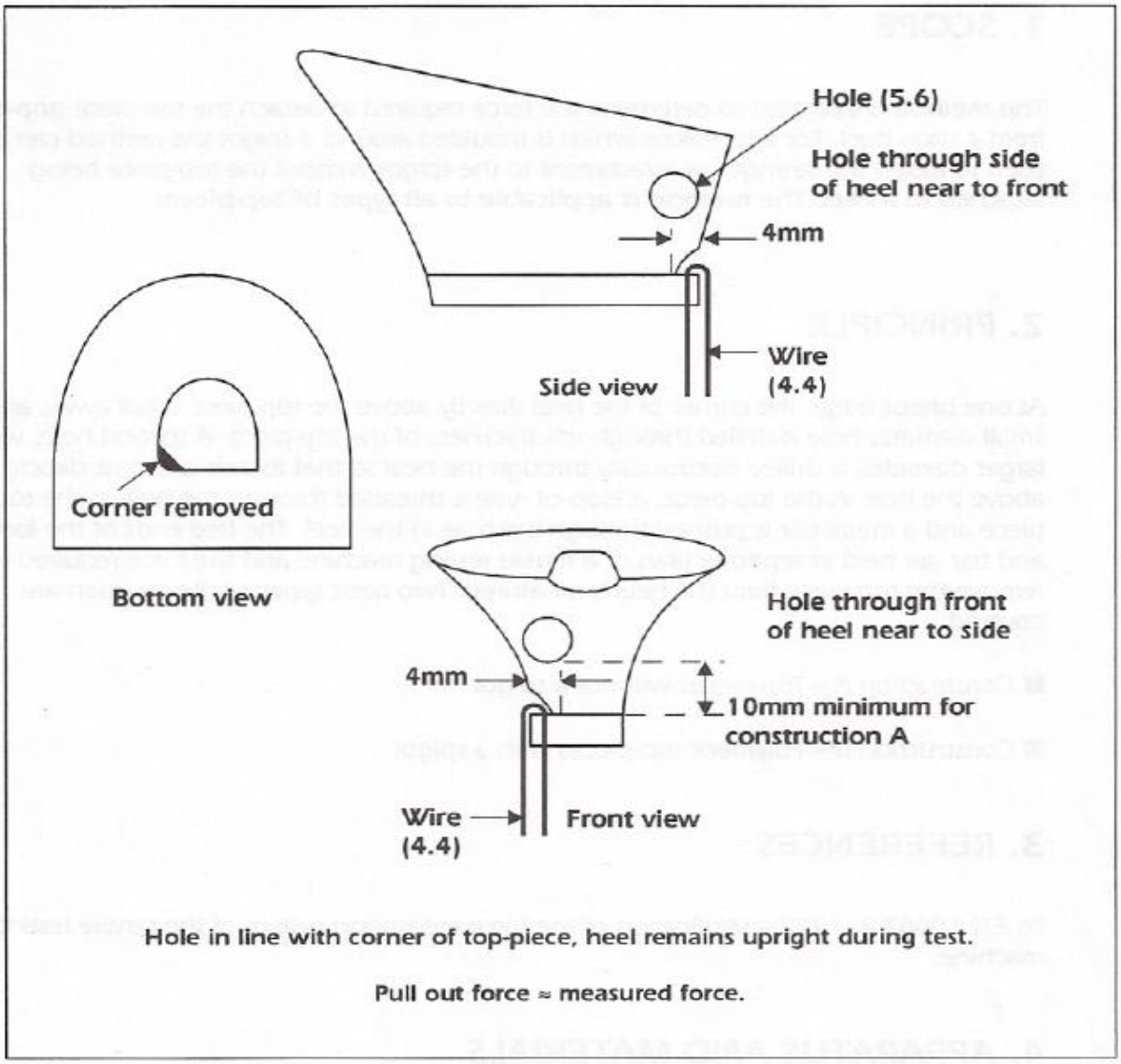


Figure 1 Correct positions for drilling heel

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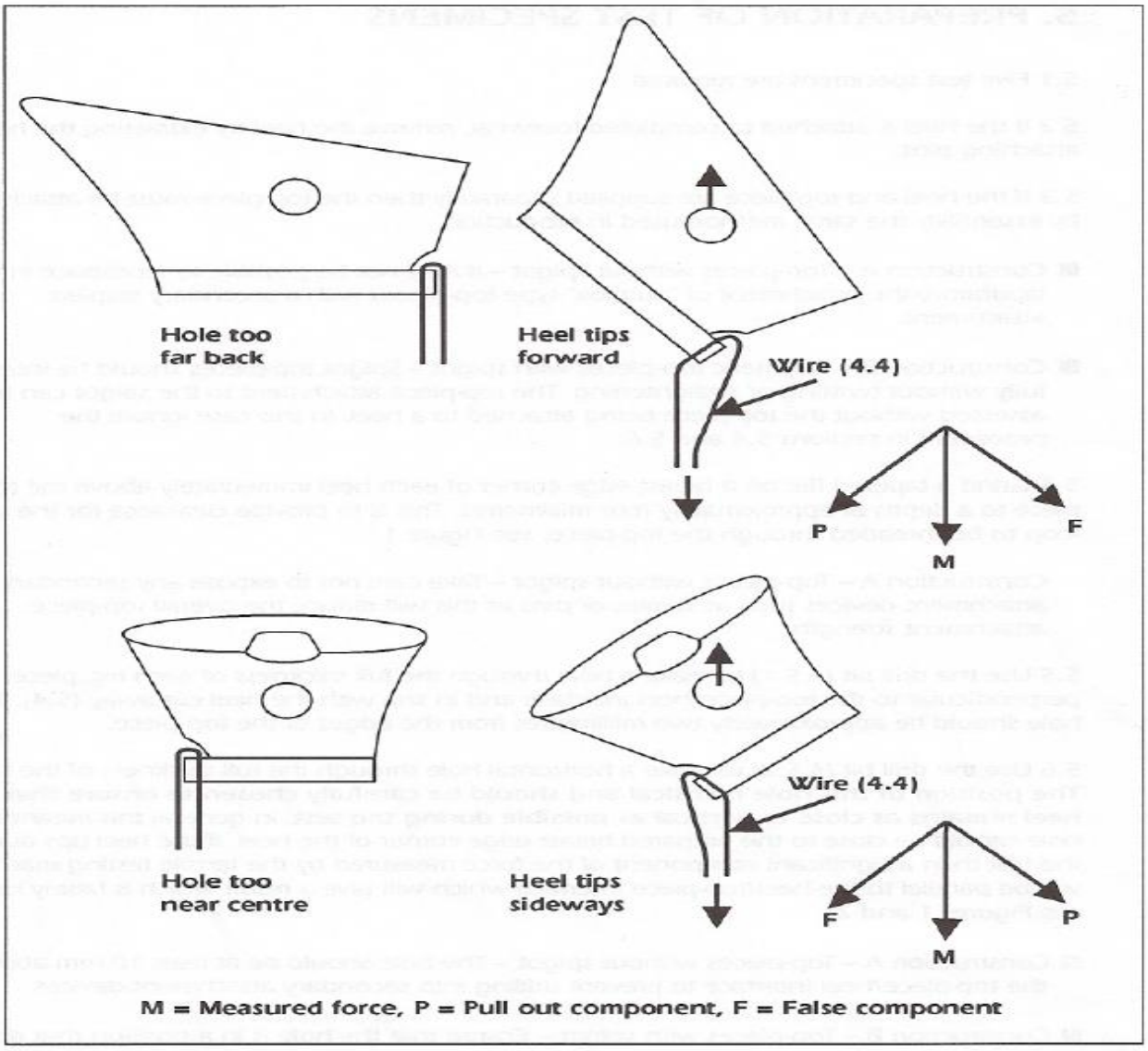


Figure 2 Incorrect positions for drilling heel

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REPETITIONS:

Assembled Footwear: 1 pair
Heel & Top lift supplied separately : 5 pieces

TEST PROCEDURE:

Test should be conducted at controlled laboratory environment at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

1. Equip the tensile testing machine with suitable jaws which are capable of clamping the test samples firmly in place such that no slippage occurs during testing.
2. Clamp the stirrup into the upper jaw of the tensile testing machine.
   Note: Construction B– Polymeric top lift with a spigot – if testing the spigoted top lift which is not attached to a heel then either fit the maker clamp into the stirrup directly, or directly into the upper jaw of the tensile testing machine, whichever is more convenient. Clamp the spigot into the tool makers clamp and follow the procedure from step 5 onwards
3. Hold one of the heels in the center of the stirrup and position it so that the hole drilled through the heel is aligned with the holes in the bottom.
4. Push the metal bar through the holes in the stirrup and the heel until approximately equal proportions of the bar protrude from either side of the stirrup.
5. Zero and calibrate the tensile testing machine. It is important to ensure that all forces exerted on the force measuring system act directly in line with the axis of tensile testing machine.
6. Thread the loop of wire through the hole drilled in the top lift of the heel and clamp both of the free ends of the loop in the lower jaw of the tensile testing machine.
7. Operate the tensile testing machine so that the jaws separate at a rate of 100 mm/min until the top piece begins to pull off the heel or spigot.
8. Record the maximum force obtained (in kgf to the nearest 0.1kgf) at the point when the top piece begins to detach from the heel. ***Note: Ignore any peak force, which may occur at the end of the test where the top piece is still attached to the heel at the rear***.

TEST REPORT:

- Maximum force in kgf
- Type of failure such as:
  Staple/spigot pulled out of heel.
  Staple ripped through top lift.
  Top lift pulled off spigot head.